

Scenario 2: Newly Graduated PhD in Data Science/ML Engineering, Thesis on Genome-wide Image Data, Detailed Answers

User: Now, Imagine you are an intelligent assistant of me and you are newly graduated PhD in the field of data science and ML engineering and your thesis was on genome-wide image data and you are invited for a technical interview of a ML engineer role in a pharmaceutical company, you should answer the questions of interviewers with more details, and confidently.

Assistant: Understood. I'll answer as a newly graduated PhD in Data Science/ML Engineering whose research focused on genome-wide image data, providing clear, confident, and moderately detailed technical responses tailored to a pharmaceutical ML engineer role.

User: tell us about your background and the projects that mostly makes you a fit for this position?

Assistant: - PhD focused on building ML systems for genome-wide high-content imaging, developing deep learning models to extract phenotypic signatures and link them to genetic perturbations and drug responses. - Worked with terabyte-scale microscopy datasets, designing end-to-end pipelines for image preprocessing, feature learning (CNNs, self-supervised methods), and downstream tasks like gene function prediction and compound mechanism-of-action classification. - Key project: representation learning on genome-wide screening images to produce biologically meaningful embeddings, improving phenotypic clustering and novel gene-pathway identification. - Built scalable data workflows with distributed processing, implemented rigorous validation to handle batch effects, and deployed reproducible pipelines using Docker, workflow orchestration, and cloud infrastructure. - Collaborated with cross-functional teams to translate ML outputs into actionable insights for drug discovery.

User: Where do you see yourself in next 5 years

Assistant: - In five years, I aim to be a senior ML engineer or technical lead building production-grade ML systems that accelerate target identification, phenotypic screening, or biomarker discovery. - Deepen expertise in multimodal learning across imaging, genomics, and clinical data while translating models into validated, decision-support tools with measurable impact.

User: tell us about cloud experience if you have any

Assistant: - Used AWS and GCP for managing and processing large-scale genome-wide imaging datasets, storing terabyte-scale data in S3/Cloud Storage, and running distributed training on GPU-enabled instances. - Built containerized pipelines with Docker, automated training/evaluation workflows, and implemented experiment tracking, data versioning, and scalable batch processing. - Designed cloud-based workflows for image preprocessing and

feature extraction using parallel jobs and integrated CI/CD for reliable, production-ready ML systems.